

Script-Controlled ACL – Restrict Record Access Based on Field Value

Team ID : SWTID-2026-1756

Team Size : 4

Team Leader : VINITHA J

Team member : REVATHI M

Team member : JENCY R

Team member : BABY SHALINI S

- **Script-Controlled ACL – Restrict Record Access Based on Field Value**
- **Description :**
- This lab demonstrates how to implement a **script-controlled Access Control List (ACL)** to restrict users from viewing records unless they meet specific conditions. In this scenario, only users belonging to the **EEE branch** are allowed to view records where the **Branch field is set to EEE**, while administrators retain full access.
- **Timer :**
- 3 hrs
- **Milestone-1 Creation of Users and Roles**
- **User Creation:**
- **Log in** to the ServiceNow instance with **admin access**.
- **Create a test user:**
 - Navigate to **User Administration → Users**
 - Click **New**
 - **User ID:** EEE User
 - **First name:** EEE
 - **Last name:** User
 - **Email:** eeeuser@gmail.com
 - Save the record

Role Creation

- Navigate to **User Administration** → **Roles**
- Click **New**
- **Name:** Enter role name as bb1
- **Description:** Brief role purpose
- Click **Submit**
- Create other roles also like **bb2,bb3,bb4.(follow the same procedure)**
- Assign all the roles to the EEE User

Tables Creation

1. In the **Application Navigator**, search for **System Definition** → **Tables**.
2. Click **New** and create a custom table with the following details:
 - **Label:** Institution Details
 - **Name:** u_institution_details
 - **Extends for :** false

1. Click **Submit**.
2. Add the following fields to the **u_institution_details** table:
 - **Student Roll Number**– Auto Number
 - **Student Name** – Reference- User
 - **Faculty Name** – Reference- User
 - **Branch** – Choice (ECE, EEE, CSE)
 - **Email**– String
 - **Phone Number**– String
 - **Description**– Multi String
3. Save the Table

1. Create multiple records in the **Student Records** table with different branch values (ECE, EEE, CSE).

Creation of Access Control Lists(Read ACL)

- 1. In the **Application Navigator**, search for **System Security → Access Control (ACL)**.
- 2. Elevate “Security_admin” role
- 3. Click **New** to create a new ACL.
- 4. Enter the following details:
 - **Type:** record
 - **Operation:** read
 - **Name:** u_institution_details
 - **Active:** true
 - **Advanced:** true
- 5. In the **Requires role** related list, add:
 - bb1 (custom role)
- 6. Add Data condition as **Branch is EEE**
- 1. In the **Script** field, add the following script:

```
(function () {  
  
    // Allow admin users full access  
    if (gs.hasRole('admin')) {  
        return true;  
    }  
    // Allow only EEE branch users to see EEE records  
    if (gs.hasRole('bb1')){  
        return true;  
    }  
    // Deny access for all others  
    return false;  
})());
```

- 2. Click **Submit** to save the ACL.

1. Impersonate **EEE users** and open the Student Records list.
2. Copy the table name u_institution_details.list/LIST and paste in the All menu.
3. Impersonate **Admin** and open the Student Records list.

Verify :

- User with **bb1 role** can view **only EEE branch records**
- User without the role cannot view any records
 - Admin user can view **all records regardless of branch**

Creation of Access Control Lists(Create ACL)

1. Click **New** to create a new ACL(Create)
2. Enter the following details:
 - **Type:** record
 - **Operation:** create
 - **Name:** u_institution_details
 - **Active:** true
3. In the **Requires role** related list, add:
 - bb2 (custom role)
4. No need to add any Data condition
5. Save the ACL Record

Verify :

- User with **bb1 &bb2 roles** can view **only EEE branch records b New** Button

Creation of Access Control Lists(Write ACL)

1. Click New to create a new ACL(Write)
2. Enter the following details:
 - Type: record
 - Operation: Write
 - Name: u_institution_details
 - Active: true
3. In the Requires role related list, add:
 - bb3 (custom role)
4. No need to add any Data condition
5. Save the ACL Record

Verify :

- User with **bb1 &bb2 &bb3** roles can view and edit the **EEE branch records along with New Button**

Creation of Access Control Lists(Delete ACL)

1. Click **New** to create a new ACL(Delete)
2. Enter the following details:
 - **Type:** record
 - **Operation:** Delete
 - **Name:** u_institution_details
 - **Active:** true
3. In the **Requires role** related list, add:
 - bb4 (custom role)
4. No need to add any Data condition
5. Save the ACL Record

Verify :

- Users with **bb1 &bb2 &bb3&bb4** **roles** can see, create and edit along with delete the **EEE branch records**.

Outcome :

By completing this lab, learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE, and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation, and deletion across forms, lists, and Service Portal interfaces.